# Performative Prediction in a Single-Asset Market

*Self-fulfilment and signal erosion under forecast adoption*

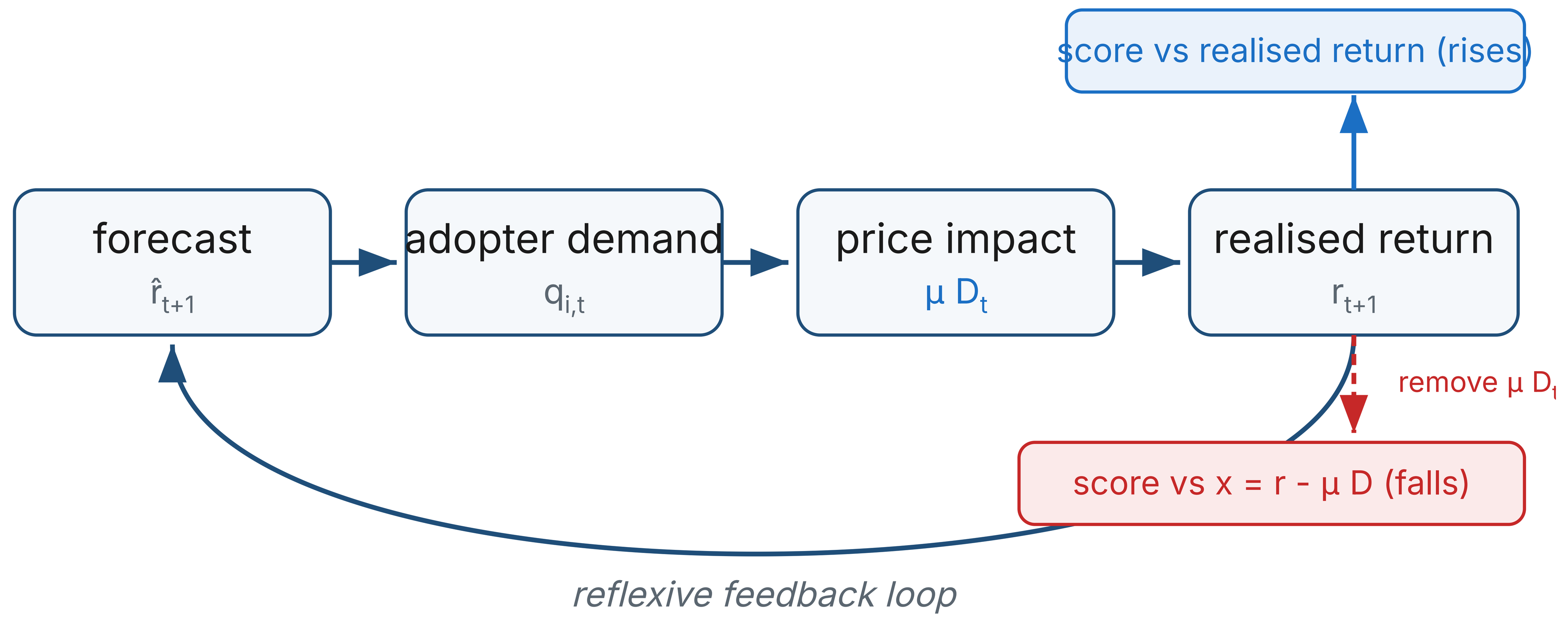
Jan Jacek Wejchert | Supervisor: Nacho Molina Clemente
IE School of Science and Technology | MSc in Business Analytics and Data Science | July 2026

## One forecast, two verdicts, as adoption rises

**0.07 → 0.19** realised-return $R^2$ (rises)

**0.08 → 0.04** demand-adjusted $R^2$ (falls)

**0.28 → 0.45** effective persistence φ (rises)




$$r_{t+1} = \mu D_t + x_{t+1}$$

$\mu D_t$: self-fulfilment $x_{t+1}$: independent signal

One coordinated order flow, scored two ways. The forecast moves the price it is judged against; stripping that move out leaves the signal that was truly there.

## The question

A published forecast acquires users. Once adopters trade on it, their coordinated demand enters the very returns the forecast is scored against. So does adoption make a forecast look better, worse, or both at once? We separate the score a deployed forecaster can measure from the signal that is independently present.

## The model

A minimal single-asset market. Traders hold mean-variance demand with constant absolute risk aversion (Brock and Hommes 1998); a risk-neutral market maker absorbs net order flow and moves the quote (Beja and Goldman 1980; Farmer and Joshi 2002).

$$r_{t+1} = \varphi\, r_t + \mu D_t + \sigma\, \varepsilon_{t+1}$$

$$p_{t+1} = p_t + \mu D_t + \sigma\, \varepsilon_{t+1}$$

$$z_{i,t} = \frac{\hat{r}_{i,t+1}}{a\,\sigma^2}$$

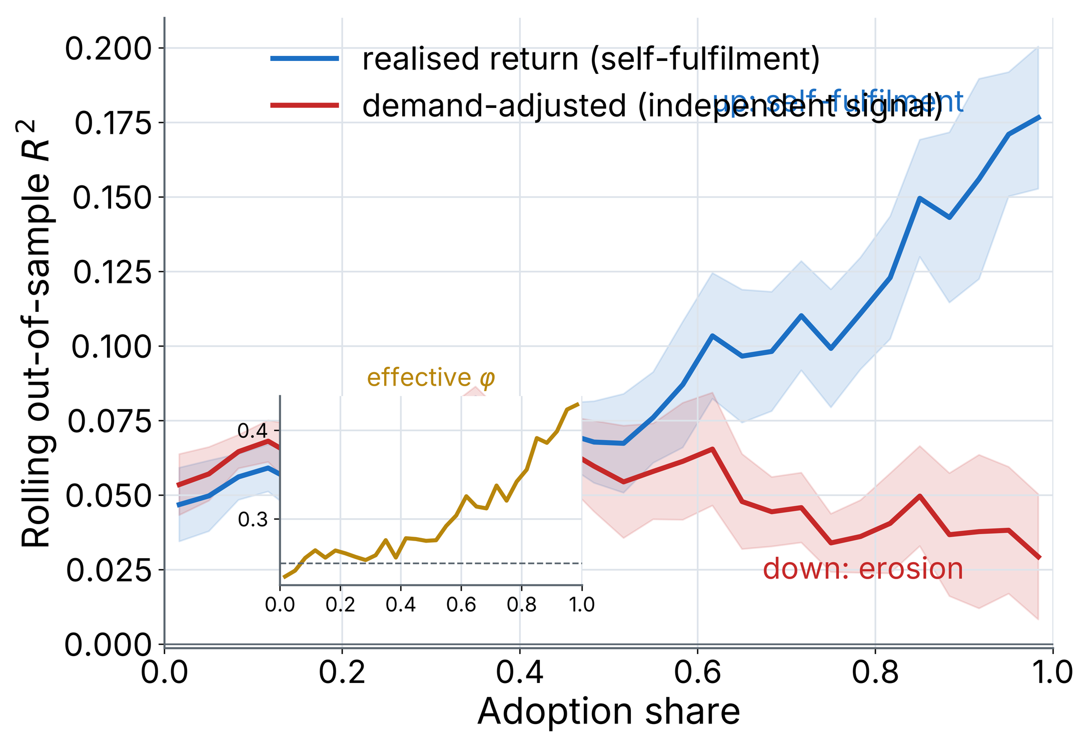
Capped mean-variance demand sets each adopter order from the shared forecast; aggregate demand $D_t$ clears through the maker, who carries the inventory.

## Intra-period timing

1. Agents observe returns through t and form forecasts.
2. Adopters and non-adopters submit orders.
3. The maker absorbs demand and moves the quote.
4. The news shock and the persistence term realise.

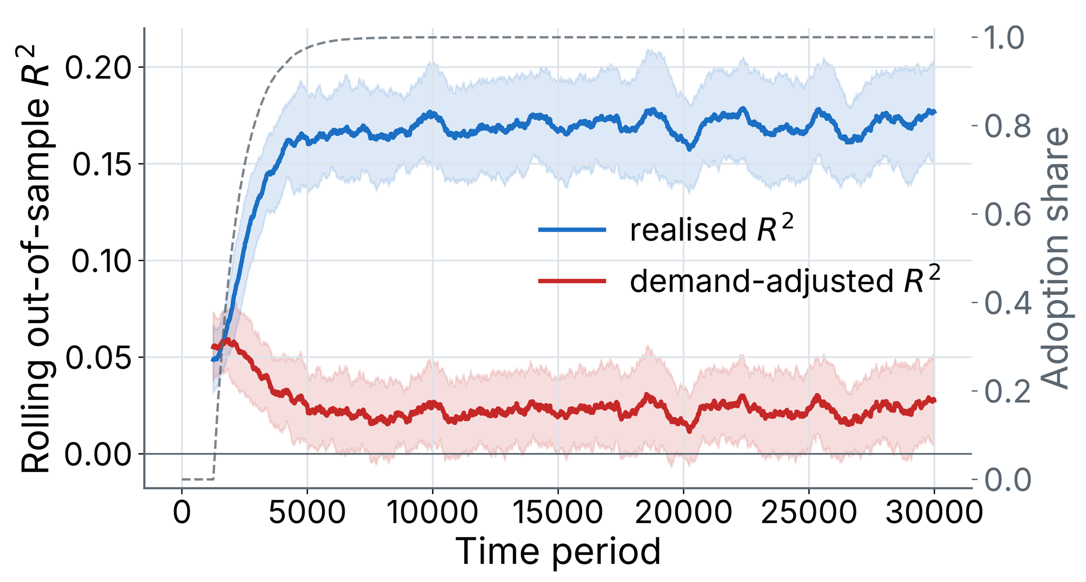
## Two evaluation targets

The same forecast is scored against two targets. The realised return $r_{t+1}$ is what a deployed forecaster sees. The demand-adjusted return $x_{t+1} = r_{t+1} - \mu D_t$ is a counterfactual that removes contemporaneous price impact: the independent signal. Effective persistence φ links the two channels.



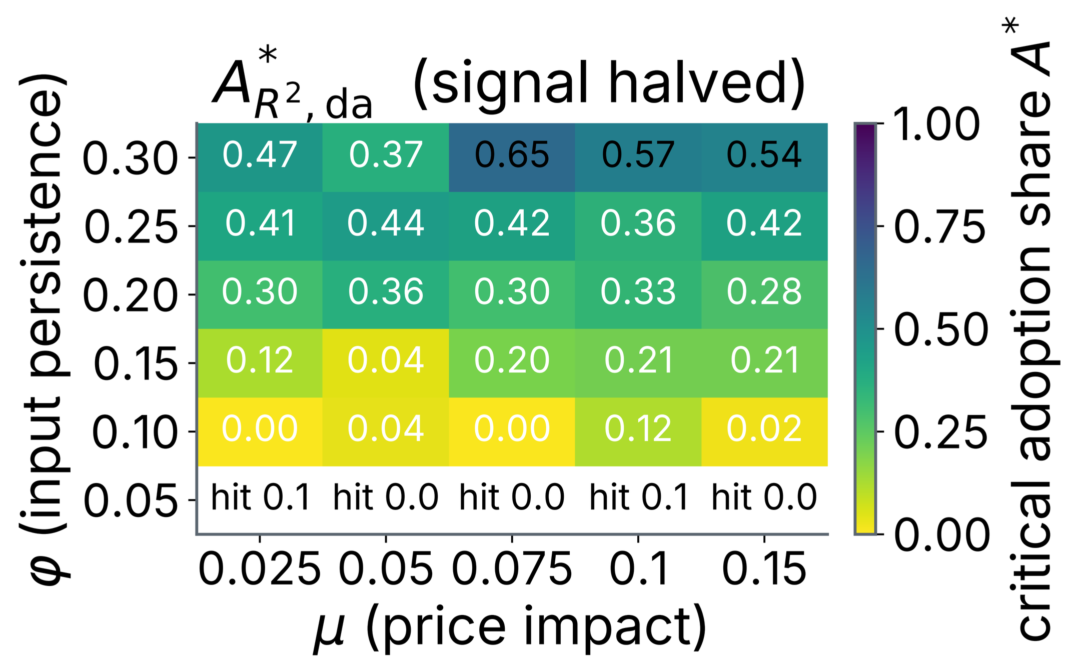

Rolling out-of-sample $R^2$ against adoption share, 100 seeds, ±1 SD bands. Realised (blue) rises by self-fulfilment; demand-adjusted (red) erodes on the same path.

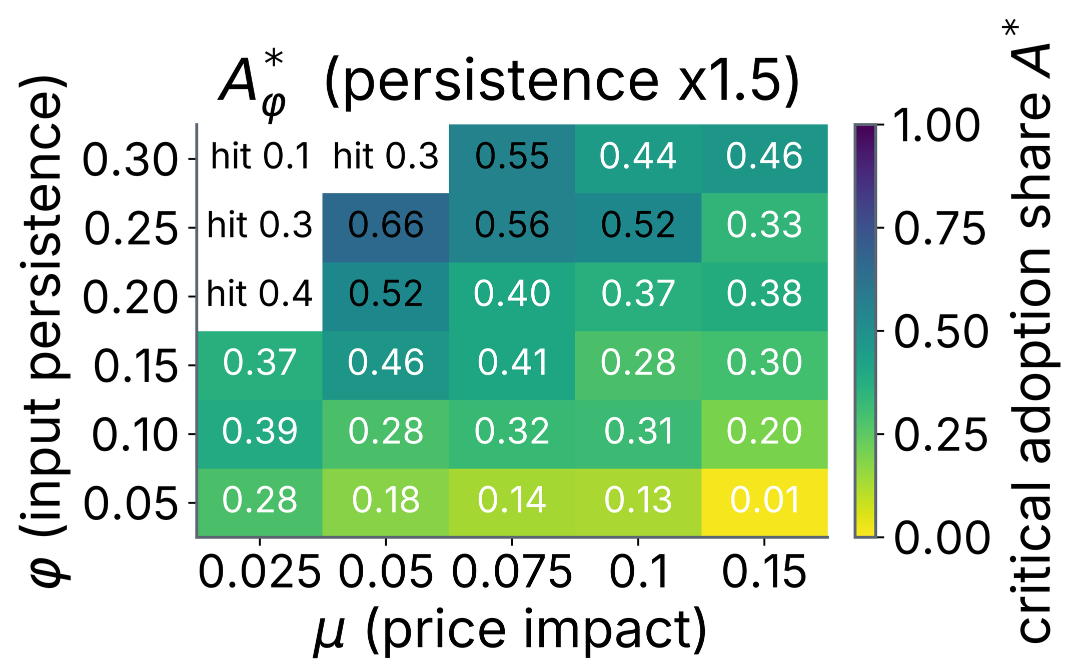
## A steady state, not a transient




Holding high adoption to T = 30000 over 50 seeds, the readings hold separated plateaus (realised near 0.17, demand-adjusted near 0.02). The gap is structural, not an artefact of the adoption ramp.

## Where the signal erodes




A 3600-run sweep over (μ, φ), window, AR order p in {1, 2, 5, 10}, 10 seeds. The demand-adjusted threshold $A^*_{R^2,da,rel}$ is reached in 67 of 90 AR(1) cells; the persistence threshold $A^*_{\varphi,rel}$ in 82 of 90. The realised channel reaches its threshold in only 28 of 90, all weak-signal noise crossings: the designed non-result. The pattern holds across p.

## Conclusion

Realised-return $R^2$, the only score a deployed forecaster can compute, over-reports skill once the forecast has users with price impact. Report both readings together; their gap measures the deployment-driven component of measured accuracy.

## Future work

The economic endpoint (proportional costs, net trading value against a null benchmark, the profit threshold) is deferred: a faithful reading must first neutralise the position-size asymmetry between the capped adopter and uncapped null rules. A heterogeneous trader ecology is likewise left for future work.

Soros 1987 · Perdomo et al. 2020 · Brock & Hommes 1998
Beja & Goldman 1980 · Farmer & Joshi 2002 · Diebold & Mariano 1995

**Take-home.** Realised $R^2$ over-reports skill once a forecast has users with price impact. Report both readings; their gap is the deployment-driven component of measured accuracy.

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